Whose Perception of Geopolitical Risk Is Priced in Defence Equities?

Evidence from the Russia–Ukraine War

Khrystyna Kateryna Valenia

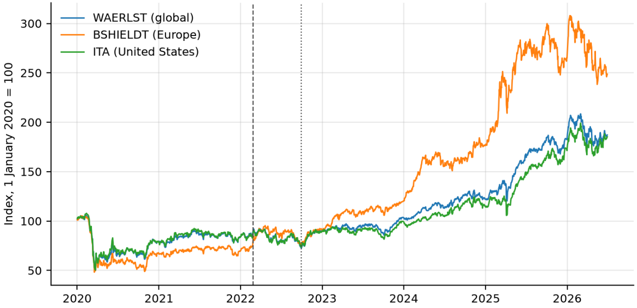
Master's thesis defence, August 2026

Roadmap

- **Why this topic** — the repricing to be explained, and what the literature leaves open
- **Research question and contribution**
- **Data sources and sample** — why the obvious corpus will not do
- **Identification strategy and model** — a conditional joint test with a positive control
- **Results** — four tests, two asset classes, one non-market outcome, and out of sample
- **Findings** — what a null of this kind does and does not license
- **Robustness** — why this null is a finding rather than an absence
- **Conclusions**

The repricing to be explained

- Europe **+21.6%** in the five weeks from 24 February 2022; **+7.0%** globally, **+9.1%** in the US.
- Then a second and larger step: Europe **+63.7%** in the first half of 2025, on rearmament budgets.
- **What information drove that repricing?**



Defence-equity index levels, 1 January 2020 = 100. Dashed line: 24 February 2022.

What the literature establishes — and the gap

- **Risk measured from newspaper text.** Baker et al. (2016) and Caldara and Iacoviello (2022) established normalised newspaper counts as a workable proxy. Both are built from a fixed panel of English-language, mostly American and British outlets.
- **Media content moves prices.** Tetlock (2007) at short horizons; Manela and Moreira (2017) at long ones.
- **Defence equities and conflict are more equivocal than the summaries suggest.** Event studies find sharp responses at onsets — Covachev and Fazakas (2024) up to 12%, Klomp (2025) 10–15%. But continuously measured geopolitical risk predicts defence *volatility*, not returns: Apergis et al. (2018), Zhang et al. (2023).
- **The closest paper.** Bondarenko et al. (2024): for the Russian domestic economy, local Russian-language risk measures carry information that English-language measures do not.
- **The gap.** No study separates news by the nationality of the outlet that published it. So when a defence-equity price is said to respond to geopolitical risk, it is responding to a quantity built from **one editorial vantage point** — and that vantage point is an untested modelling assumption.

Research question and contribution

- **Whose perception of geopolitical risk is priced in defence equities?**
- Two candidate answers:
  - *Either* the information originates where the war is fought — local media are closer, cover it earlier and far more heavily — and reaches Western prices with a delay.
  - *Or* local coverage adds nothing once the Western signal is already in the regression.
- Only the second is directly testable, because it is a statement about **incremental** information. Establishing the positive claim that Western coverage is itself priced would need a different design, and is not claimed here.
- **Contribution.** (i) The mirror of Bondarenko et al. (2024), run on the counterparty's assets. (ii) The editorial vantage point made the object of study rather than an assumption. (iii) A methodological point: at corpus level, this class of measurement error is invisible in the output. (iv) A refinement of the defence-equity literature — the response is specific to the Western reporting layer.

The measurement problem that dictates the data

- Comparing national media ecosystems requires a corpus that actually contains them. The stream most readily available for this — GDELT's Global Knowledge Graph 1.0 — does not.
- A representative daily file holds **60,690 records**. Of these, **7** carry a .ru domain and **21** a .ua domain.
- Nationality in that stream can be assigned only from the countries an article *mentions*, which for **88.6%** of articles is not the country that published it.
- A series built that way and labelled Ukrainian sentiment therefore measures the tone of **English-language writing about Ukraine** — one editorial population sorted by topic and relabelled as three perspectives. Any comparison between them compares two topic proxies, not two vantage points.
- **The fix:** GDELT's translingual archive — **65 source languages**, machine-translated through a single annotation pipeline, with each article identified by **the outlet that published it**.

Five ecosystems, classified by publisher

- **UA, RU_STATE, RU_INDEP, WEST, EN_GLOBAL.** An article in a British newspaper about Russian troop movements is Western, not Russian.
- Four tiers in priority order: a hand-curated register, then national domain suffix, then source language conditional on country, then a residual category.
- **Country beats language at every tier.** Ukrainian outlets publish heavily in Russian — 24tv.ua carries 2,595 Ukrainian-language and 1,865 Russian-language articles. A language-first rule would file those as Russian and manufacture agreement between two ecosystems the design must keep apart.
- **One rule, pointing both ways.** State-funded external broadcasters classify to the funding state, which puts Deutsche Welle and RFE/RL in the Western block; exile newsrooms classify to their country of origin, which keeps Meduza in the Russian independent one.
- **Validated externally, not asserted.** Each registered domain is resolved against Wikidata: of 84 outlets, 66 resolve and **63 agree — precision 0.955**. All three disagreements are exile newsrooms, the one case where origin and operation genuinely differ.

Variables, targets and sample

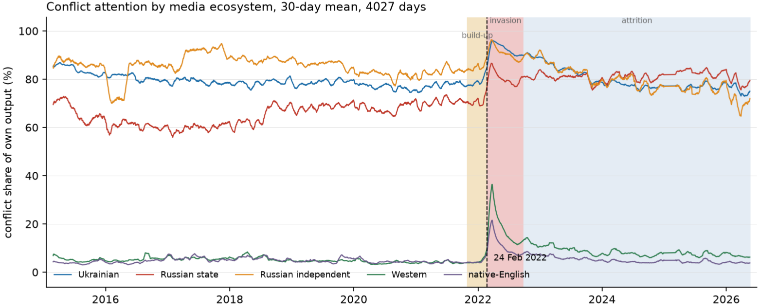
- **Two perception indices per ecosystem per day.** *Ukraine/Russia coverage share* — articles whose GDELT location field names a place in Ukraine or Russia, over the ecosystem's **entire** daily output. *Coverage tone* — mean GDELT tone of those same articles.
- **A third, theme-based split** following Caldara and Iacoviello (2022): anticipatory coverage of threats and build-ups against realised coverage of acts and attacks. Conflict-specific by construction.
- **Shares, never raw counts.** GDELT source coverage drifts by a factor of two and a half over eleven years; a count series would show that drift as a trend in every ecosystem at once.
- **Changes, never levels.** In levels every series carries a common war-regime trend that the financial controls already capture.
- **Five targets:** Bloomberg WAERLST (global) and BSHIELDT (Europe) from 2020; the ITA ETF back to 2015; and two hand-built equal-weighted baskets.
- **Sample: 4,027 days, 18 February 2015 to 20 May 2026, 98% of calendar days**; 3,073 carry all five ecosystem series. Supplementary: Ukrainian Air Force records — 3,812 attack records, 102,396 weapons launched, **74.3%** intercepted.

Identification strategy and model

- $r_{i,t} = \alpha + \beta_{West} X_t^{West} + \beta_{Local} X_t^{Local} + \gamma Z_t + \varepsilon_{i,t}$
- The object of interest is a **joint F-test on β_{Local}, with the Western block already in the regression.** That conditioning is what separates local media adding something from local and Western media correlating with returns for the same reason.
- **The positive control is the identical test on β_{West}, in the same cells.** If the design detects neither block, a null on the local block means nothing. This is what makes the result a finding.
- **The market control must be regional.** European defence returns are controlled with the STOXX 600, not the S&P 500. Over the build-up the two markets correlate only 0.409, so a US control leaves ordinary European variation in the residual — and that residual correlates 0.26 with the threat measure being tested.
- Two controls only, plus HAC(5) standard errors throughout. The grid crosses two frequencies, six episode windows plus the full sample, and five targets: 70 cells, of which **31** clear a minimum-observation rule.
- **31 tests at 5% would produce about 1.5 false rejections by chance**, so Benjamini–Hochberg correction is applied across all 31 at a 5% false discovery rate. Gates 3, 4 and 5 are **pre-registered**; **Gate 2 is not**, and is reported as the exploratory test it is.

Descriptive: two media worlds, an order of magnitude apart

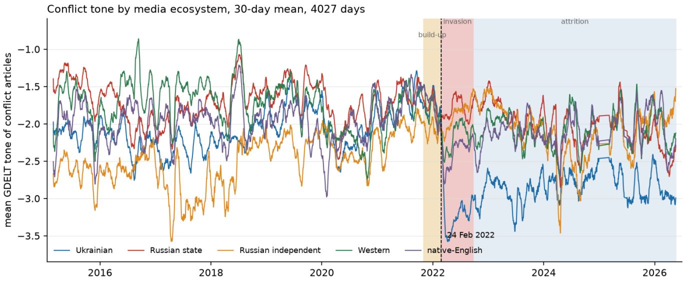
- Local outlets: **71–85%** of output names a Ukrainian or Russian place. Western: **7.2%**. Native-English: **5.4%**.
- At the invasion Western attention moves **19.0% to 40.9%** in a day — Ukrainian only 90.3% to 95.8%.
- **The local ecosystems had almost no headroom left to react.**



Ukraine/Russia coverage share by ecosystem, 30-day moving averages, 2015–2026.

Descriptive: the tone asymmetry at the invasion

- Tone shift at the invasion: Ukrainian -1.66, native-English -0.64, Western -0.36 — **Russian state** $+0.02$.
- **Not a composition artefact:** on a fixed panel of the 24 state outlets present on both sides, the shift is -0.05.
- **The signal that moved most sharply at the defining event of the sample is not the one prices track.**



Mean coverage tone by ecosystem, 30-day moving averages, 2015–2026.

Gates 2 and 3: local perception and defence-equity returns

Block and news timing	Cells	BH surv.	Min. p	Verdict
Gate 2 — coverage volume and tone (exploratory)				
Local, lagged one day (primary)	31	1	9.0×10^{-4}	not robust
Local, same-day	31	7	2.2×10^{-6}	not tradeable
Western, lagged one day (control)	31	0	2.7×10^{-3}	not detected
Gate 3 — anticipated/realised split (pre-registered)				
Local, lagged one day (primary arm)	31	2	5.5×10^{-6}	**FAIL**
Western, lagged one day (control)	31	2	1.6×10^{-4}	**detected**

Notes: Joint F-tests on one block, with the other block and the controls already in the regression. Benjamini–Hochberg at 5% FDR across all 31 cells of an arm.

- **The timing contrast is the sensitivity evidence.** Local coverage is detected in **7 of 31** cells when credited to its publication day, and in **1 of 31** when lagged one trading day so that all of it precedes the open — and that one cell is the 2017–19 window on ITA at weekly frequency, years before the full-scale invasion.
- **Gate 3 fails both pre-registered arms.** The build-up window produces no survivor (smallest p 0.018), and the two that do survive share a window where the rule required two. Both are weekly cells fitting 13 parameters to 58 observations, while the grid's two deepest cells — 2,754 observations — return $p = 0.08$ and 0.36.
- **The Western block is detected in the same 31 cells, under the same correction.** That asymmetry is what the test was built to isolate.

Gates 4 and 5: gas and escalation

Gate and window	n	p (local)	p (Western)
Gate 4 — Dutch TTF natural gas returns			
(a) Build-up and invasion, daily	222	0.399	**0.00004**
(b) Shutdown and aftermath, daily	231	0.533	0.193
(c) Full crisis, daily	563	0.250	0.066
(c) Full crisis, weekly	133	0.907	0.052
Gate 5 — realised escalation, held-out sample			
Horizon $h = 1$ day	637	0.212	0.171
Horizon $h = 5$ days	629	0.227	**0.033**

Notes: Both gates were pre-registered before their data were ingested. All four Gate 4 cells estimated are reported.

- These answer the fair objection that **defence equities are a weak testbed** — the link from a Russian newspaper to a US contractor's share price runs entirely through Western investors.
- **Gas has a direct physical channel:** Russia supplied roughly 40% of EU gas, and TTF went from about 20 euro to over 300. Yet **all four pre-registered conditions fail** — no surviving cell; the Brent placebo breached at $p = 0.0014$; 0.399 becoming 0.840 without the ten largest TTF moves; and the ordering inverted. **Western detected at $p = 4 \times 10^{-5}$.**
- **Escalation removes prices entirely.** On 637 held-out days local media do not lead realised conflict. The weakest of the four gates: its own Western control is significant at five days, not at one.

Out of sample — and what a null can bear

True R^2_{OS} implanted	Long sample (1,855 days, 3 targets)	Bloomberg (686 days, 2 targets)
0.0% (nominal size)	0.02	0.02
0.2%	0.40	0.21
0.5%	**0.81**	**0.44**
1.0%	0.98	0.76
2.0%	1.00	0.97

Notes: Simulated power: rejection rate at the 5% level, 1,000 paths per cell, so each entry carries a standard error of at most 1.6 percentage points.

- 50 specifications — five targets by ten predictors — in an expanding window, one day ahead. Every comparison is **nested**, so it is arbitrated by **Clark–West, not Diebold–Mariano**: under nesting DM has no standard normal limiting distribution and is biased against the larger model even when its extra predictor carries real information.
- **Zero rejections, against 2.5 expected by chance.** Best R^2_{OS} observed **+0.10%**; smallest p-value 0.067, so not one specification reaches even nominal significance.
- **The power simulation is what makes that informative — and it must be read as two numbers, never one.** On the three long-sample targets an effect of 0.5% is detected in **81%** of paths, and one of 1.0% in **98%**. On the two Bloomberg targets the same effects give **44%** and **76%**, so there a meaningful effect would be missed more often than not, and the null is weak evidence rather than a bound.
- **Power is not exclusion.** 81% means one sample in five misses a real effect.

What this means

- **Local-language perception adds nothing to defence-equity returns conditional on Western coverage** — in coverage volume, in tone, in the anticipated/realised structure, across two asset classes, against one non-market outcome, and out of sample. The null is consistent across all four tests.
- **This is a finding rather than an absence, because the design demonstrably detects media effects.** The Western block survives correction in the pre-registered anticipation test and is detected at $p = 4 \times 10^{-5}$ in gas; the local block itself is detected in 7 of 31 cells before the tradeable lag is imposed. The informative feature is the **co-occurrence** — a null on the local block in precisely the cells where an effect is found on the Western one.
- **The descriptive asymmetry points the same way.** Russian state tone did not move at the invasion and local attention was already at saturation, while Western attention doubled in a day. If the marginal buyers of Western defence equities form their views from Western coverage, they responded to the one series that registered the event.
- **The mirror of Bondarenko et al. (2024) comes back reversed — but the symmetric positive claim is not made.** This design identifies the local block's incremental contribution, not the Western block's own effect. What the two studies jointly support is narrower and more useful: **the informative ecosystem is the one inhabited by the agents whose decisions are being measured.**

Robustness: the tests that could have overturned this

- **The classification rule.** Five rules run through the full grid — baseline, register-only, no language tier, language-first, and aggregators retained. The verdict is unchanged under all five: one or two survivors of 31. Language-first is the informative failure: it **cannot represent a Russian independent block at all**, because the language tier claims those articles before ownership is consulted.
- **Aggregation.** An index-level null could mask cross-sectional variation. A firm-level panel of **31 defence names and 85,065 firm-days**, with exposure from published SIPRI arms revenue, produces no exposure gradient in any war window — nominal significance appears only in the years *before* the full-scale invasion, and **0 of 10 cells survive correction**.
- **The physical war itself**, in case the news measures are simply a poor proxy for it. A forecasting race over **144 specifications** and five information sets: 16 achieve a positive R^2_{OS} and **none survives correction**. Realised attack intensity is no more informative about subsequent defence returns than perceived intensity.
- **Volatility**, where a positive result was most plausible, since volatility is persistent where returns are not. Forty-eight augmented HAR-RV-X specifications: **none improves significantly**, eight are significantly worse, and all six correction survivors are deteriorations. Reported with its limit — on a squared-daily-return proxy the HAR family is roughly twice as inaccurate as GARCH, so the test has limited power.
- Both forecasting exercises are **exploratory, not pre-registered**, and are labelled that way.

Six results that dissolved — and why that matters

The result, when it was significant	What eliminated it
Defence volatility responds to threat	The regional market control (S&P 500 to STOXX 600)
Threat moves returns in the build-up, $p = 0.0001$	The same control: p becomes 0.84
Gate 3 clears its pre-registered rule, 7 survivors	Adding the held-out window: 7 becomes 2, verdict FAIL
Local perception priced in gas, $p = 0.0028$ on 81 days	Replication on 222 days of the same period: p becomes 0.399
Local media anticipate escalation, both halves	The pre-registered held-out sample: p becomes 0.21
A state-versus-independent censorship wedge	Correcting the outlet register: $p = 0.561$

Notes: Each correction to the register moved the last one *further* from significance. None of the six is claimed as a result anywhere in the thesis.

- Each was significant at conventional levels, each admitted a plausible economic mechanism, and each survived at least one robustness check before failing another. **The six failure modes are all different** — two omitted variables, one truncated sample, one small sample, one in-sample split that did not generalise, one composition change.
- **In every case the eliminating test is one that a study reporting the positive would have had no particular reason to run.** That is what bounds how much confidence a single significant specification warrants in this literature.
- They are reported rather than suppressed because **in a literature whose regressors are constructed from text, the set of choices under which a result survives is part of the result** — and a null established against this background is considerably more informative than one obtained without it.

Conclusions

- **The answer.** Local coverage provides no robust incremental information about defence-equity returns beyond what Western coverage already carries. It is a null, not a reversal — and it is not attributable to a design unable to detect media effects.
- **The implication beyond this conflict.** Multilingual geopolitical-risk indicators are built on the premise that sources closer to an event carry information Anglophone coverage omits. Tested in close to the most favourable setting available — eleven years, the most intensively covered conflict in modern media history, an asset class whose valuation rests on it, local media at near-saturation coverage — **the premise does not hold at either horizon at which the information could have been traded.** It is not refuted in general. It cannot be assumed.
- **Limitations, stated as bounds.** The coverage-share measure is close to a domestic-news filter for local outlets, so part of the null may be a ceiling — bounded by Gate 3, which is theme-based and returns the same null, and by local daily changes being two to four times as variable as Western ones. The design is associational, not causal. GDELT tone is dictionary-based. And daily aggregates cannot resolve intraday timing; only timestamped data could settle what the Gate 2 contrast means.
- **Extensions.** Other conflicts — Israel–Hamas, the Taiwan Strait — to separate a general property of defence-equity pricing from a feature of this war. Intraday prices, at which the anticipated-versus-realised distinction becomes observable. And assets whose marginal investors are **not** Western, which is the sharpest available test of the proposition that each market prices the information ecosystem its own participants inhabit.

Backup slides

Held for questions.

The main specification, estimated

	(1) Controls	(2) +Western	(3) +local, same-day	(4) +local, lagged
Δ Ukrainian attention			0.009	-0.011
Δ Russian state attention			-0.001	-0.001
Δ Russian indep. attention			-0.125^{***}	0.010
Δ Ukrainian tone			-0.073^{*}	-0.038
Δ Russian state tone			0.127^{**}	-0.055
Δ Russian indep. tone			0.033	-0.017
STOXX 600 return	0.925^{***}	0.923^{***}	0.927^{***}	0.921^{***}
Log VIX (lagged)	0.068	0.066	0.075	0.066
Observations	921	921	921	921
R^2	0.348	0.351	0.365	0.354
p, local block jointly zero	—	—	**0.003**	0.734
p, Western block jointly zero	—	0.333	0.688	0.177

Notes: Dependent variable: daily percentage return on BSHIELDT, over the 921 days on which it trades and all five news series exist. News regressors are standardised first differences, so a coefficient is the return change per one-standard-deviation move. Western-block coefficients omitted for space; none is significant. Newey–West HAC(5). Stars: 1%, 5%, 10%.

- Column (3) against column (4) is the whole story in one cell: the local block is jointly significant same-day at $p = 0.003$, and jointly indistinguishable from zero at $p = 0.734$ once lagged by one trading day.

Does the classification rule drive the result?

- Five rules are run through the full 31-cell grid: the **baseline**; **register-only**, dropping the ccTLD and language tiers; **no language tier**; **language-first**; and one **retaining news aggregators**.
- **The local-block verdict is unchanged under all five** — one or two survivors of 31 under the primary alignment in every variant.
- **Language-first is the informative failure.** Because the language tier claims Russian-language articles before ownership is consulted, it produces **no Russian independent block at all**. That is the concrete cost of getting the country-before-language rule wrong, and it is why the rule is stated as a rule rather than applied case by case.
- Three outlets were in fact misfiled under the state-broadcaster rule before it was written down — which is why that rule is stated in both directions.

Why the regional market control is decisive

- Controlling European defence returns with the S&P 500 is superficially reasonable — the two markets are highly correlated over long horizons.
- **Over the build-up window their correlation is only 0.409.** A US control therefore leaves a substantial component of ordinary European market variation in the residual.
- **That residual correlates 0.26 with the threat measure being tested**, so the omitted regional factor loads onto the variable of interest.
- The consequence: a build-up threat effect significant at $p = 0.0001$ under the S&P 500 control becomes $p = 0.84$ under the STOXX 600. Two of the six dissolved results share this single cause.
- Every regression in the thesis therefore uses the regional market return appropriate to its target: STOXX 600 for European targets, S&P 500 for global and US ones.

Why Clark–West and not Diebold–Mariano

- Forecast-accuracy comparisons in this literature are most often reported with the **Diebold–Mariano** statistic (1995). It is not valid here.
- Each of the 50 specifications compares the historical mean **augmented with one perception predictor** against the historical mean alone. Setting that predictor's coefficient to zero recovers the benchmark exactly, so **the benchmark is nested within the model**.
- Under nesting, DM has **no standard normal limiting distribution** and is **biased against the larger model** even when its additional predictor carries genuine information — so a DM-based null here would be partly an artefact of the statistic.
- **Clark and West (2007)** correct for the estimation noise that nesting introduces, giving a statistic that is valid in this setting and one-sided in the direction of improvement.
- Diebold–Mariano with the Harvey et al. (1997) small-sample correction is retained for genuinely non-nested comparisons. In the code the two are separate functions, so the distinction cannot be flagged away.

Return forecasting race, by information set

Set	Contents	Specs	$R^2_{OS} > 0$	Best R^2_{OS}	BH surv.
F	Financial controls only	24	1	0.019	0
P	F + physical attack variables	24	4	0.073	0
N	F + news variables	24	1	0.033	0
PN	F + physical + news	24	3	0.060	0
PNG	PN + pairwise ecosystem tone gaps	24	4	0.059	0
	Historical mean	12	0	0.000	—
	AR(1)	12	1	0.009	—

Notes: Expanding-window one-step-ahead forecasts on WAERLST, BSHIELDT and ITA, at horizons of one and five days, on the full sample and on the September 2022 to June 2026 attack subsample. Ridge and gradient boosting, 144 rows in total. Exploratory, not pre-registered.

Volatility forecasting race, HAR-RV-X against HAR-RV

Model	Specs	Mean QLIKE gain (%)	Sig. better	Sig. worse	BH surv.
HAR-RV-X (N)	12	-4.4	0	1	0
HAR-RV-X (P)	12	-65.0	0	2	2
HAR-RV-X (PN)	12	-62.6	0	2	2
HAR-RV-X (PNG)	12	-64.8	0	3	2

Notes: Mean QLIKE by model class, lower is better: GARCH 0.95, GJR-GARCH 0.96, EGARCH 0.98, **HAR-RV 1.93**. That last figure is the limitation, not an aside: realised variance here is the sum of squared daily returns, because intraday prices were not available, so the HAR family is roughly twice as inaccurate as the GARCH family it is normally competitive with.

- Augmenting a volatility model with war variables **degrades** its accuracy on this measure rather than improving it: of 48 specifications none improves significantly, eight are significantly worse, and all six correction survivors are deteriorations.
- What **cannot** be concluded is that conflict news carries no information about defence-equity volatility. A test built on squared daily returns has limited power to detect it.

Does the index-level null hide a cross-section?

- The objection is fair: an index-level null could mask cross-sectional variation, with heavily war-exposed firms responding where a diversified index does not.
- **A firm-level panel of 31 defence names and 85,065 firm-days**, with exposure measured from published SIPRI arms-revenue data, tests it directly.
- **No exposure gradient appears in any war window.** Nominal significance appears only in the years before the full-scale invasion, and in a full sample of which 59% falls in those years.
- **0 of 10 cells survive Benjamini–Hochberg correction.** The index-level result is therefore not an artefact of aggregation.
- Firm-level constituent and SIPRI exposure data could not be extended further, which is why the thesis carries no cross-sectional chapter and says so.